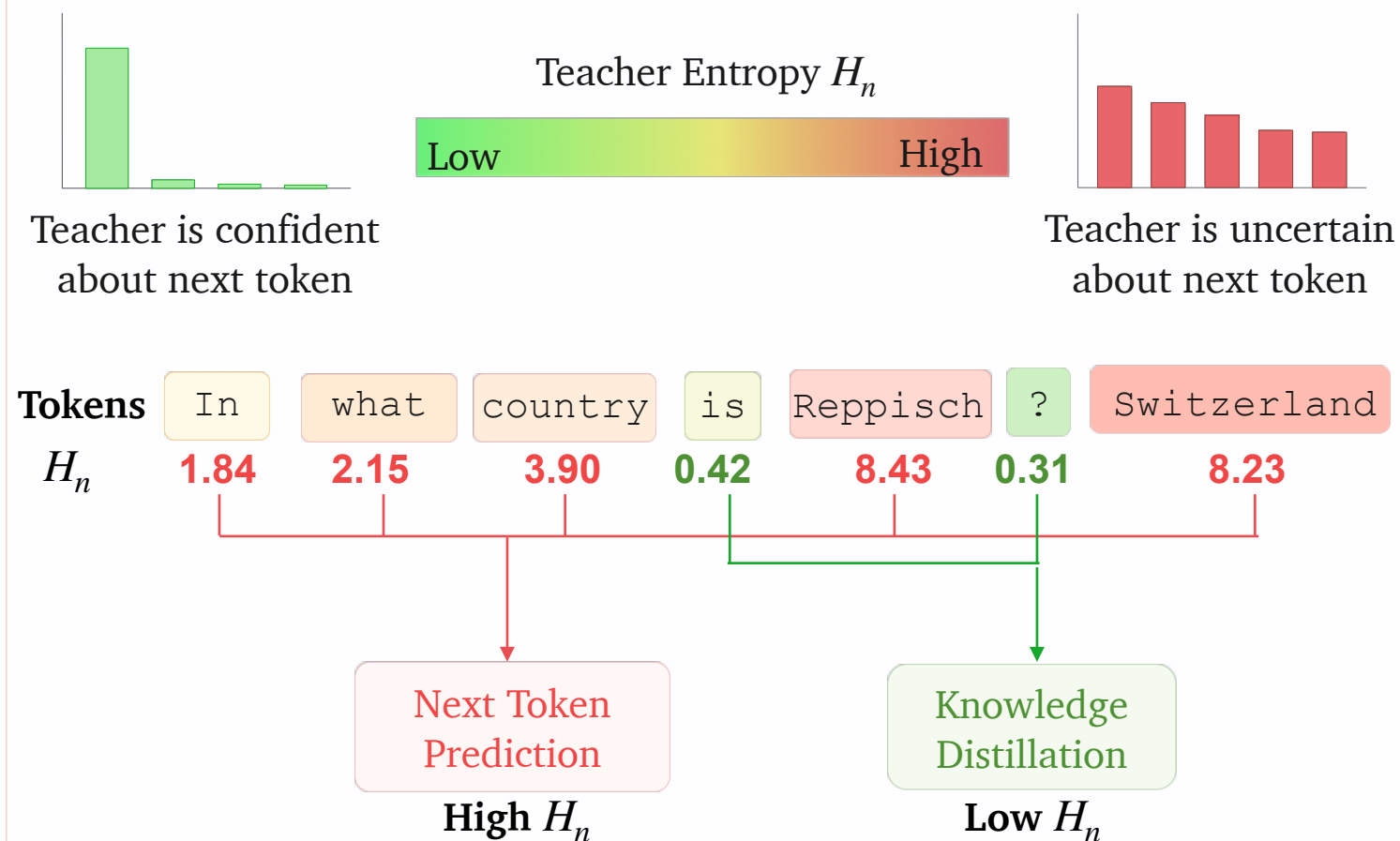


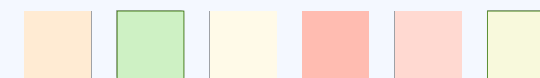
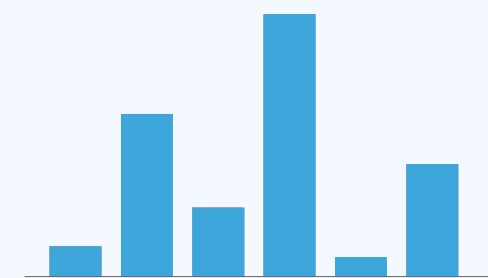
Motivation: Teacher entropy varies by token



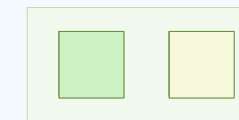
Intuition: Distill from the teacher when it's confident; rely on corpus when it's uncertain.

Switch Distillation (SD) at every training step

- 1 Compute per-token entropy**
From teacher logits scaled with temperature τ , compute per-token entropy:
$$H_n = - \sum_v p_n^\tau(v) \log p_n^\tau(v)$$
- 2 Select lowest $q\%$ tokens**
Within the batch, select the $q\%$ of tokens with the lowest H_n .
- 3 Route tokens to losses**
 $q\%$ of tokens $\rightarrow$ reverse KD
 $(1-q\%)$ of tokens $\rightarrow$ CE
- 4 Optimize student model**
Backpropagate the combined loss to update the student.

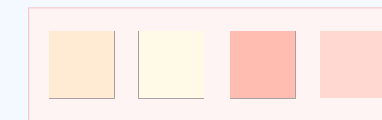


Reverse KD



$q\%$

Cross-Entropy



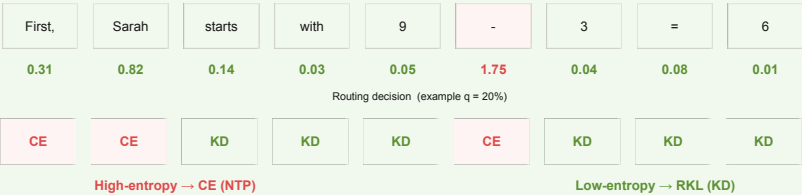
$1 - q\%$



(b) SwitchDist routes tokens by entropy (within the same sequences)

Compute entropy per token, then take the lowest-q% across the entire batch to route to RKL; others to CE.

Procedural reasoning example (same as left)



Factual query example (same as left)

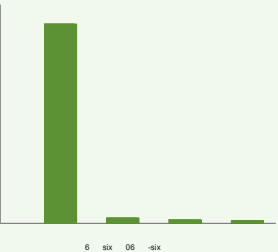


Across the entire batch (all supervised tokens):
rank by entropy H_t → select lowest q% → RKL (KD)
remaining (1 - q)% → CE (NTP)

(c) Why it makes sense: confident vs. uncertain teacher predictions

Low-entropy token (e.g., “6” in the reasoning example)

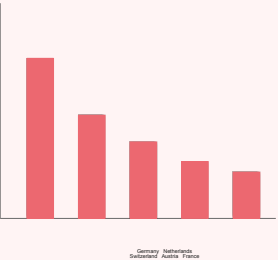
$H_t = 0.0246$
(confident)



- Mass is concentrated on one continuation.
 - Teacher is very confident.
- Use Reverse KL (KD):
distill from teacher.

High-entropy token (e.g., “country” in the factual query)

$H_t = 3.90$
(uncertain)

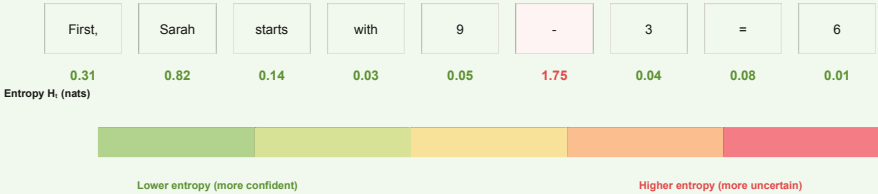


- Mass is spread over many plausible continuations.
 - Teacher is uncertain.
- Use CE (NTP): learn from the corpus instead.

Entropy is a proxy for teacher reliability on each token.

Procedural reasoning (mostly low entropy)

Prefix: Hmm, let’s break it down. First, Sarah starts with 9 apples and eats 3, so she has $9 - 3 =$
Ground-truth token: 6



Factual query (high-entropy tokens appear in the middle)

Prefix: In what country is Reppisch?
Ground-truth token: Switzerland

